# Phoenix OS — RISC OS 5 'SBPr' Directory Format

## Reverse-Engineered Reference — April 2026

**Project:** Phoenix OS (bare-metal AArch64 RISC OS-inspired kernel for Raspberry Pi 4B)
**Discovered:** Boot 228–232, April 2026
**Author:** R Andrews
**Status:** Confirmed working — boot233 decodes all 9 root directory entries

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## 1. Background

During development of the Phoenix OS FileCore driver, the root directory of a Lexar USB stick formatted on a CM4-based RISC OS 5 machine was found to use a directory format with magic bytes `'SBPr'` rather than the classic `'Hugo'/'Nick'` format documented in the RISC OS PRM.

The `'SBPr'` format was reverse-engineered entirely from hex dumps captured via Phoenix OS UART logging (boots 228–232) and cross-referenced against DiscKnight reports.

**Disc details at time of discovery:**

- Device: Lexar USB Flash Drive, 8GB
- Disc name: `laxarusb` (note: intentional spelling)
- RISC OS version: RISC OS 5.xx (CM4 build)
- Filesystem: SCSIFS (FileCore new-map, ADFS G / big-disc format)
- log2bpmb=10, bpmb=1024, secperlfau=2, nzones=1924, id_len=19

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## 2. Format Overview

The `SBPr` directory is a **2048-byte block** (4 × 512-byte sectors) divided into three regions:

| Region | Offset | Size | Description |
|---|---|---|---|
| Header | `0x000` | 64 bytes | Fixed-size metadata block |
| Body | `0x040` | 1920 bytes | Variable-length directory entries |
| Tail | `0x7C0` | 64 bytes | SIN index + sequence check |

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## 3. Header Format (bytes 0x000–0x03F)

| Offset | Size | Field | Example | Notes |
|---|---|---|---|---|
| `0x00` | 1 | `seq_counter` | `0x09` | Incremented on each directory write. Matched at tail `0x7FC`. |
| `0x01-03` | 3 | reserved | `00 00 00` | Padding |
| `0x04-07` | 4 | `magic` | `'SBPr'` | `0x53 0x42 0x50 0x72` — directory format identifier |
| `0x08-0B` | 4 | `version` | `0x00000001` | Format version = 1 |
| `0x0C-0F` | 4 | `dir_size` | `0x00000800` | Directory size in bytes = 2048 |

| | | | | |
|---|---|---|---|---|
| `0x10–13` | 4 | `entry_count` | `0x00000009` | Number of entries in this directory |
| `0x14–17` | 4 | `name_bytes` | `0x0000005C` | Total bytes of all filenames concatenated |
| `0x18–1B` | 4 | `self_sin` | `0x02FAD601` | SIN (Sector Indirect Number) of this directory |
| `0x1C–1F` | 4 | `parent_ref` | `0x00000D24` | Reference to parent directory (TBC) |
| `0x20–23` | 4 | `flags` | `0xFFFFFF5C` | Observed value — semantics TBC |
| `0x24–27` | 4 | `checksum` | `0x9C2BF6F8` | CRC or hash of directory contents |
| `0x28–2B` | 4 | `dir_size2` | `0x00000800` | Directory size repeated |
| `0x2C+` | — | entries | — | Body entries start immediately |

**Raw hex (confirmed from boot232):**

```
0x000: 09 00 00 00  53 42 50 72  01 00 00 00  00 08 00 00
0x010: 09 00 00 00  5c 00 00 00  01 d6 fa 02  24 0d 00 00
0x020: 5c ff ff ff  f8 f6 2b 9c  00 08 00 00  01 d7 fa 02
0x030: 08 00 00 00  05 00 00 00  00 00 00 00  5c ff ff ff
```

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## 4. Entry Format (body, bytes 0x040–0x7BF)

Entries are **variable length** and **4-byte aligned**. They start at offset `0x2C` (immediately after the 44-byte fixed header, before the first entry the remaining header bytes `0x2C–0x3F` may contain entry data).

| Offset | Size | Field | Notes |
|---|---|---|---|
| `+0x00–03` | 4 | `sin` | SIN of this object. Bit 0 = 1 for new-map (always set for ADFS G). Identifies files and directories equally — directories have their own `SIN` that points to another directory block. |
| `+0x04–07` | 4 | `file_size` | File size in bytes. For directories, this is typically 0 or the directory block size. |
| `+0x08` | 1 | `name_len` | Length of the name string in bytes. |
| `+0x09..` | N | `name` | ASCII filename, **not NUL-terminated**. May contain RISC OS `!` prefix for application directories. |

**Alignment:** The total entry size is `9 + name_len` rounded up to the next multiple of 4.

```
entry_size = (9 + name_len + 3) & ~3
```

**Example — `!Boot` entry at offset 0x2C:**

```
01 d7 fa 02   SIN = 0x02FAD701  ($.!Boot)
08 00 00 00   size = 8
05            name_len = 5
21 42 6f 6f 74  name = '!Boot'
[pad: 2 bytes]
```

## 5. Tail Format (bytes 0x7C0–0x7FF)

The tail provides a **SIN index** — a quick-lookup array of SINs for all entries, avoiding a full body scan.

| Offset | Size | Field | Notes |
|---|---|---|---|
| `0x7C0–0x7CF` | 16 | padding | Zeroed |
| `0x7D0–0x7D3` | 4 | padding | Zero |
| `0x7D4+` | 4×N | `sin_index[]` | One `uint32_t` SIN per entry, in the same order as body entries |
| `0x7F8–0x7FB` | 4 | `tail_name_frag` | Last 4 bytes of last entry's name (or padding) |
| `0x7FC` | 1 | `seq_counter` | Copy of header `seq_counter` — consistency check |
| `0x7FD–0x7FF` | 3 | padding/checksum | TBC |

**Confirmed SIN index (root dir, laxarusb disc):**

| Offset | Entry | SIN |
|---|---|---|
| `0x7D4` | `!Boot` | `0x02FAD701` |
| `0x7D8` | `Apps` | `0x03024001` |
| `0x7DC` | `Diversions` | `0x03047901` |
| `0x7E0` | `Documents` | `0x0304AC01` |
| `0x7E4` | `Printing` | `0x02F09401` |
| `0x7E8` | `Public` | `0x02F0DA01` |
| `0x7EC` | `Updates` | `0x02F0DB01` |
| `0x7F0` | `Utilities` | `0x02F0DD01` |
| `0x7F4` | `Wallpaper` | `0x02EFC201` |

**Raw hex (confirmed from boot232):**

```
0x7C0: 00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00
0x7D0: 00 00 00 00  01 d7 fa 02  01 40 02 03  01 79 04 03
0x7E0: 01 ac 04 03  01 94 f0 02  01 da f0 02  01 db f0 02
0x7F0: 01 dd f0 02  01 c2 ef 02  6f 76 65 6e  09 00 00 6b
```

## 6. Detection

To detect an `SBPr` directory, check bytes `[4..7]` of the first sector:

```
int is_sbpr = (buf[4]=='S' && buf[5]=='B' && buf[6]=='P' && buf[7]=='r');
```

For comparison, a classic Hugo directory has:

```
int is_hugo = (buf[1]=='H' && buf[2]=='u' && buf[3]=='g' && buf[4]=='o');
```

Note that `buf[0]` in both formats is the sequence counter (not part of the magic).

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## 7. Implementation — `adfs_dump_sbpr_dir()`

The Phoenix OS implementation in `kernel/filecore.c` (boot233+):

```
static void adfs_dump_sbpr_dir(const uint8_t *dir)
{
    uint8_t  seq         = dir[0];
    uint32_t entry_count = LE32(dir + 16);
    uint32_t self_sin    = LE32(dir + 24);

    /* Walk body entries from offset 0x2C */
    uint32_t off = 0x2Cu, found = 0u;
    while (found < entry_count && off + 9u < 0x7C0u) {
        uint32_t sin   = LE32(dir + off);
        uint32_t size  = LE32(dir + off + 4);
        uint8_t  nlen  = dir[off + 8];
        if (sin == 0u && nlen == 0u) break;

        int is_dir = (sin & 1u);
        /* print name: dir[off+9 .. off+9+nlen-1] */

        uint32_t entry_bytes   = 9u + (uint32_t)nlen;
        uint32_t entry_aligned = (entry_bytes + 3u) & ~3u;
        off += entry_aligned;
        found++;
    }
}
```

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## 8. Relationship to Classic Hugo/Nick Format

| Feature | Hugo/Nick | SBPr |
|---|---|---|
| Magic | `'Hugo'` at `buf[1..4]` | `'SBPr'` at `buf[4..7]` |
| Tail magic | `'Nick'` at `buf[2044..2047]` | Seq byte at `buf[0x7FC]` |
| Max entries | 77 (fixed 26-byte records) | Unlimited (variable-length) |
| Entry names | 10 bytes fixed | Variable (up to 255 bytes) |
| SIN index | None | Tail array at `0x7D4` |
| Directory size | 2048 bytes | 2048 bytes (same) |

| | | |
|---|---|---|
| RISC OS version | Pre-5 / all | RISC OS 5 (Pi builds) |

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## 9. Open Questions

1. **`parent_ref at [0x1C]`** : Value `0x00000D24` — may be a byte offset within the parent directory, or a parent SIN field. Needs further investigation.
2. **`flags at [0x20]`** : `0xFFFFFF5C` — likely a bitmask. The `0x5C` = `'\'` is suspicious; could be an attribute byte.
3. **Entry `file_size` for `!Boot`** : Shows `8` not `0` — application directories may store metadata here.
4. **`tail_name_frag`** : `'oven'` = last 4 chars of `'Wallpaper'`. Confirmed as name fragment, not a magic word.
5. **Load/exec addresses**: Classic Hugo entries have load/exec/length. SBPr appears to omit these from the directory (they may be stored elsewhere, or in an extended entry format not yet observed).

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## 10. Discovery History

| Boot | Finding |
|---|---|
| 184–220 | FileCore disc record read, zone map traversal, chain bugs |
| 221–227 | Chain breaks — copy2 formula bugs fixed |
| 228 | First hex dump of LBA 0x775AC8 — `SBPr` magic identified |
| 229 | Full byte scan — no `Hugo` found anywhere in 2048 bytes |
| 230 | Copy2 formula fix (`+_tnz` not `+1`) |
| 231 | Boot block cross-check — confirms `root_dir` IDA unchanged |
| **232** | **`SBPr` detection confirmed — `entries=9, self_sin=0x02FAD601`** |
| **233** | **`adfs_dump_sbpr_dir()` written — full directory listing decoded** |

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*Phoenix OS SBPr Format Reference — April 2026*
*File: `docs/SBPr_Directory_Format.pdf`*